CPU

Datei: CPU.kicad_sch

Video−Generation

Datei: Video−Generation.kicad_sch

Color−Modulator

Datei: Color−Modulator.kicad_sch

RF−Stage

Datei: RF−Stage.kicad_sch

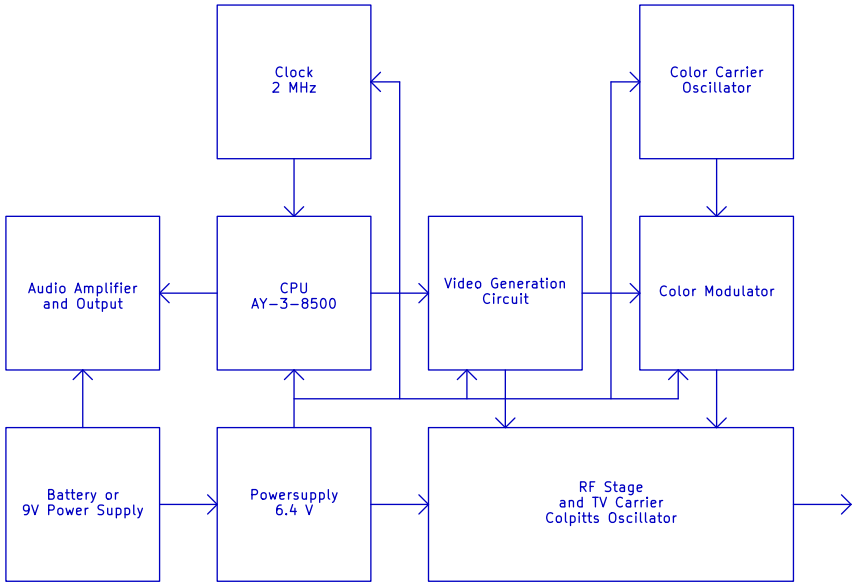
Symbol	Note
7Vrms 1	Measurement point with measured value
1	Adjustment point
1	Curve shape (oscillogram)
1	
1	Circuit Note
	Module/Cassette
	PCB
	Functional group

Note regarding component names:

All component designations written in red were either missing, illegible, or duplicated. I attempted to deduce their meaning logically and have noted below which designations this applies to.

- C1 assigned because it is unreadable
- C4 assigned because it is unreadable
- C9 assigned because there is no label
- C42 assigned because it is unreadable
- C101 assigned because it is unreadable
- C106 assigned because it is unreadable
- D21 assigned because it is duplicated (D20)
- FL1, FL2, FL3 assigned because they are unreadable
- Jx assigned because there is no label
- L101 assigned because it is unreadable
- LS1 assigned because there is no label
- Q101 assigned because it is unreadable
- RV1, RV2 assigned because there are no labels
- U1, U2, U3 assigned because they are unreadable
- XTAL1 assigned because it is unreadable

All measurements with the oscilloscope were taken with a 1:10 probe to avoid loading the RF circuits.

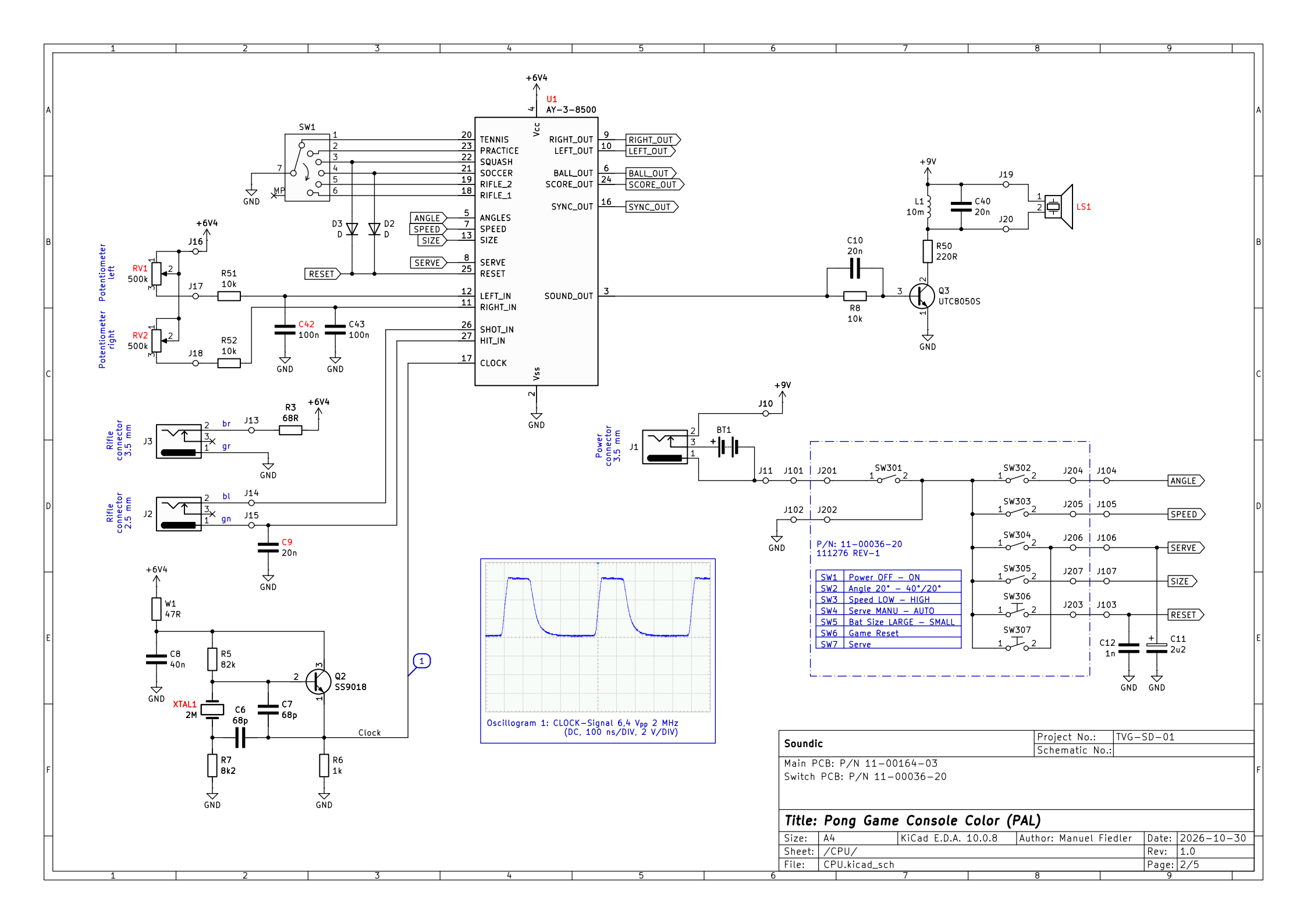


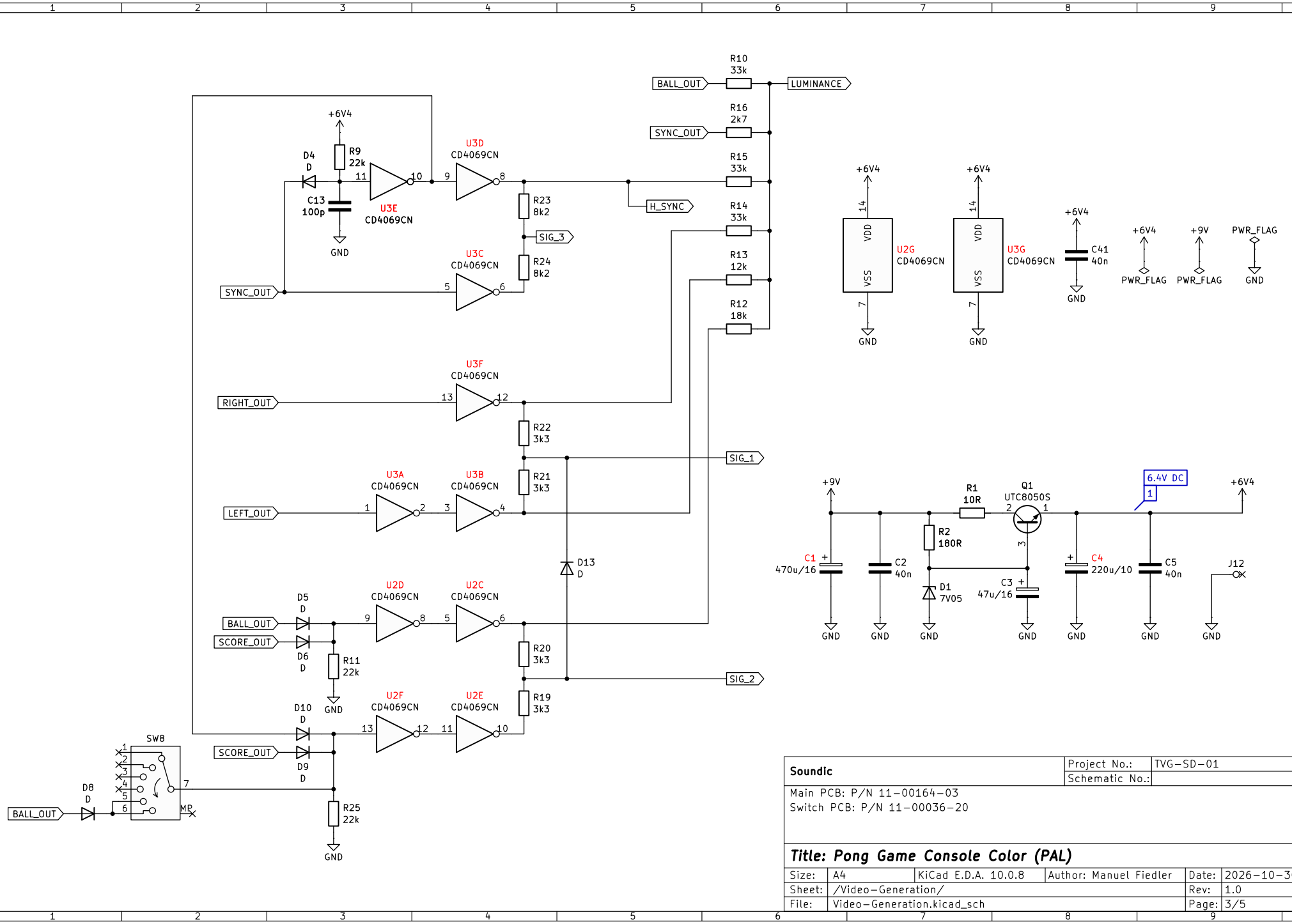
Soundic		Project No.:	TVG−SD−01
		Schematic No.:	

Main PCB: P/N 11−00164−03
Switch PCB: P/N 11−00036−20

Title: Pong Game Console Color (PAL)

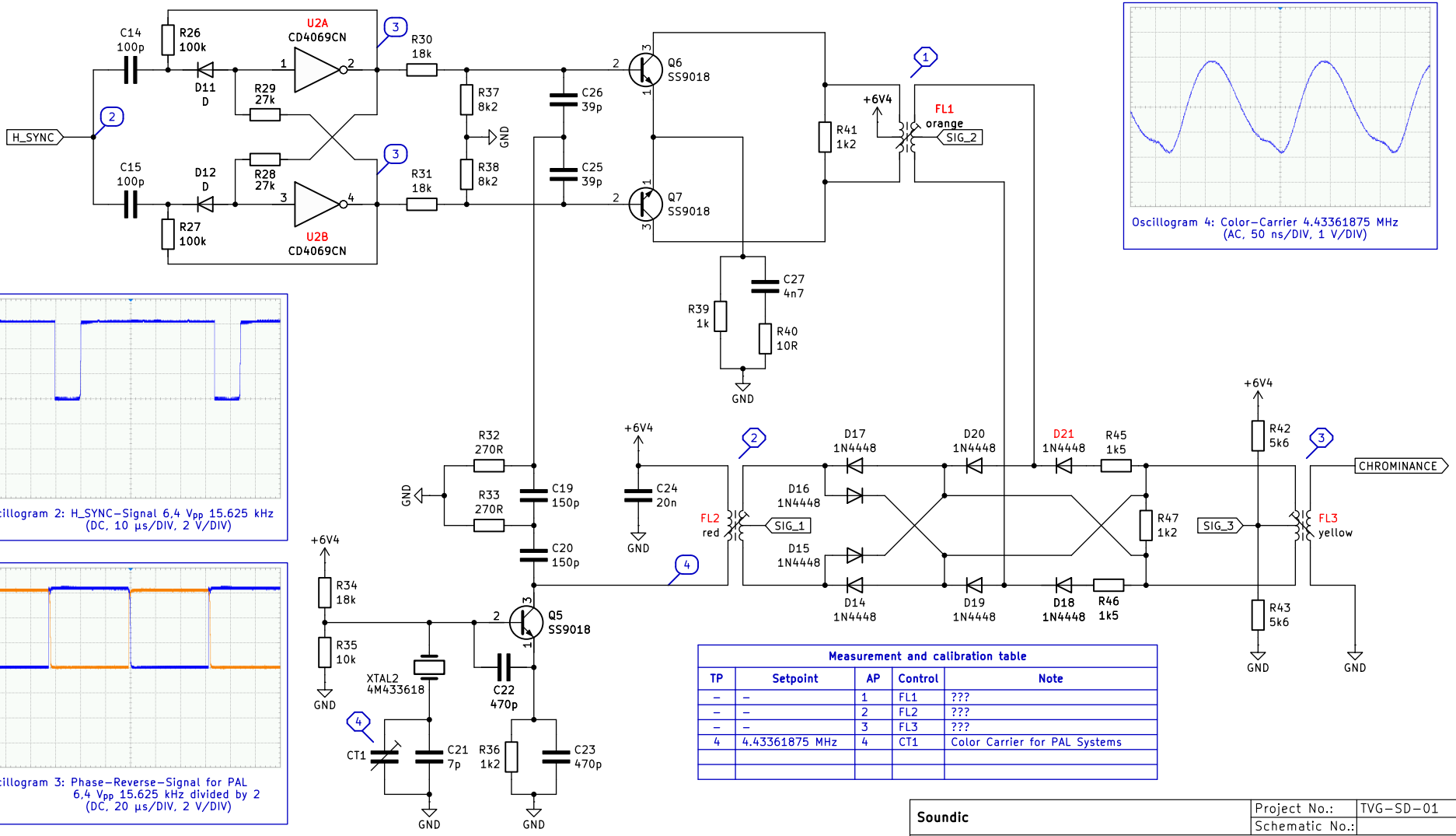
Size:	A4	KiCad E.D.A. 10.0.8	Author: Manuel Fiedler	Date:	2026−10−30
Sheet:	/			Rev:	1.0
File:	kicad_schematic.kicad_sch			Page:	1/5

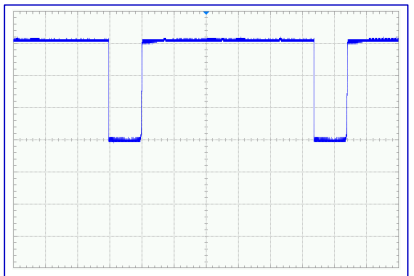

U1
AY-3-8500
+6V4
Vcc
Vss
TENNIS
PRACTICE
SQUASH
SOCCER
RIFLE_2
RIFLE_1
ANGLES
SPEED
SIZE
SERVE
RESET
LEFT_IN
RIGHT_IN
SHOT_IN
HIT_IN
CLOCK
RIGHT_OUT
LEFT_OUT
BALL_OUT
SCORE_OUT
SYNC_OUT
SOUND_OUT
SW1
RESET
D3
D2
Potentiometer left
RV1
500k
R51
10k
J16
J17
Potentiometer right
RV2
500k
R52
10k
J18
C42
100n
C43
100n
Rifle connector 3.5 mm
J3
R3
68R
J13
br
gr
Rifle connector 2.5 mm
J2
J14
J15
bl
gn
C9
20n
W1
47R
C8
40n
R5
82k
Q2
SS9018
XTAL1
2M
C6
68p
C7
68p
R7
8k2
R6
1k
Clock
C10
20n
R8
10k
Q3
UTC8050S
+9V
L1
10m
C40
20n
J19
J20
LS1
R50
220R
Power connector 3.5 mm
J1
J10
BT1
J11
J101
J102
J201
J202
SW301
SW302
SW303
SW304
SW305
SW306
SW307
J204
J205
J206
J207
J203
J104
J105
J106
J107
J103
C12
1n
C11
2u2
P/N: 11-00036-20
111276 REV-1
SW1 Power OFF – ON
SW2 Angle 20° – 40°/20°
SW3 Speed LOW – HIGH
SW4 Serve MANU – AUTO
SW5 Bat Size LARGE – SMALL
SW6 Game Reset
SW7 Serve
Oscillogram 1: CLOCK–Signal 6,4 Vpp 2 MHz
(DC, 100 ns/DIV, 2 V/DIV)
Soundic
Project No.: TVG-SD-01
Schematic No.:
Main PCB: P/N 11-00164-03
Switch PCB: P/N 11-00036-20
Title: Pong Game Console Color (PAL)
Size: A4
KiCad E.D.A. 10.0.8
Author: Manuel Fiedler
Date: 2026-10-30
Sheet: /CPU/
Rev: 1.0
File: CPU.kicad_sch
Page: 2/5



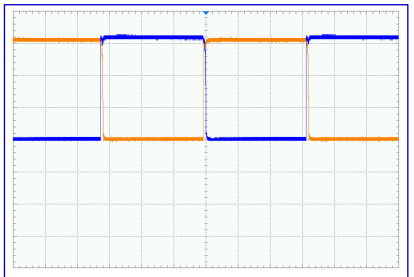
Soundic	Project No.:	TVG-SD-01	
	Schematic No.:		
Main PCB: P/N 11-00164-03 Switch PCB: P/N 11-00036-20			
Title: Pong Game Console Color (PAL)			
Size: A4	KiCad E.D.A. 10.0.8	Author: Manuel Fiedler	Date: 2026-10-30
Sheet: /Video-Generation/			Rev: 1.0
File: Video-Generation.kicad_sch			Page: 3/5

T–Flip–Flop
Frequency Divider





Oscillogram 2: H_SYNC–Signal 6,4 V_{pp} 15.625 kHz
(DC, 10 µs/DIV, 2 V/DIV)



Oscillogram 3: Phase–Reverse–Signal for PAL
6,4 V_{pp} 15.625 kHz divided by 2
(DC, 20 µs/DIV, 2 V/DIV)

Oscillogram 4: Color–Carrier 4.43361875 MHz
(AC, 50 ns/DIV, 1 V/DIV)

Measurement and calibration table

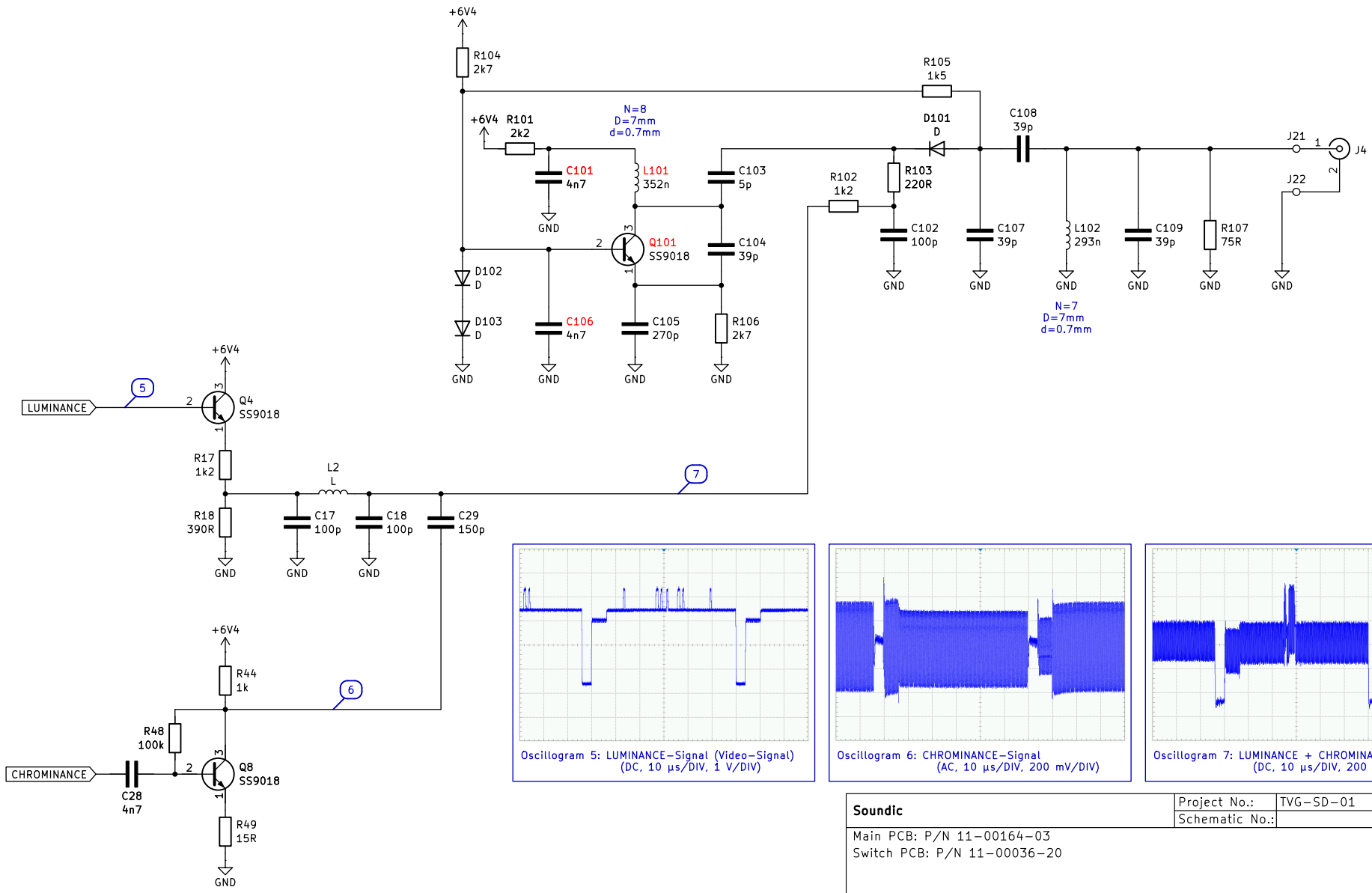
TP	Setpoint	AP	Control	Note
–	–	1	FL1	???
–	–	2	FL2	???
–	–	3	FL3	???
4	4.43361875 MHz	4	CT1	Color Carrier for PAL Systems

Soundic	Project No.:	TVG–SD–01	
	Schematic No.:		

Main PCB: P/N 11–00164–03
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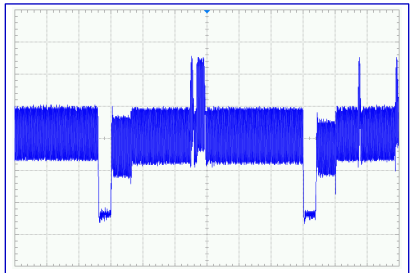
Title: Pong Game Console Color (PAL)

Size:	A4	KiCad E.D.A. 10.0.8	Author: Manuel Fiedler	Date:	2026–10–30
Sheet:	/Color–Modulator/			Rev:	1.0
File:	Color–Modulator.kicad_sch			Page:	4/5



Oscillogram 5: LUMINANCE–Signal (Video–Signal)
(DC, 10 µs/DIV, 1 V/DIV)

Oscillogram 6: CHROMINANCE–Signal
(AC, 10 µs/DIV, 200 mV/DIV)



Oscillogram 7: LUMINANCE + CHROMINANCE–Signal
(DC, 10 µs/DIV, 200 mV/DIV)

Soundic		Project No.:	TVG–SD–01
		Schematic No.:	

Main PCB: P/N 11–00164–03
Switch PCB: P/N 11–00036–20

Title: Pong Game Console Color (PAL)

Size:	A4	KiCad E.D.A. 10.0.8	Author: Manuel Fiedler	Date:	2026–10–30
Sheet:	/RF–Stage/			Rev:	1.0
File:	RF–Stage.kicad_sch			Page:	5/5